

SQL Aggregate Functions & OperatorsSection 1Question 1

SELECT DISTINCT department  
FROM Student;

| Department |
|------------|
| IT         |
| Finance    |
| HR         |

Question 2

SELECT department, AVG(age) as Avg-age  
FROM Student  
GROUP BY department

| Department | Avg-age |
|------------|---------|
| IT         | 20.5    |
| HR         | 22      |
| Finance    | 23      |

Question 3

SELECT department, COUNT(\*) as Student-count  
FROM Student  
GROUP BY department  
HAVING COUNT(\*) > 1;



| Department | Student-count |
|------------|---------------|
| IT         | 2             |
| IT         | 2             |

#### Question 4.

```
SELECT Student-id, Name, age, department
FROM Student
WHERE age BETWEEN 21 and 23;
```

| Student-id | Name    | Age | department |
|------------|---------|-----|------------|
| 2          | Bob     | 22  | HR         |
| 3          | Charlie | 21  | IT         |
| 4          | Diana   | 23  | Finance    |
| 5          | Eve     | 22  | HR         |

#### Question 5

```
SELECT Student-id, name, age, department
FROM Student
WHERE department = 'IT' or department = 'HR'
AND age > 21;
```

| Student-id | Name  | Age | department |
|------------|-------|-----|------------|
| 2          | Bob   | 22  | HR         |
| 4          | Diana | 23  | Finance    |
| 5          | Eve   | 22  | HR         |



## Section 02

Question 6

```

SELECT department, Sum(credits) as total_credits
FROM Courses
GROUP BY department
HAVING Sum(credits) > 5;

```

| Department | total_credits |
|------------|---------------|
| IT         | 11            |

Question 7

```

SELECT Course_id, course_name, department,
Credits
FROM Courses
WHERE Credits < 4;

```

| Course_id | Course_name | department | Credits |
|-----------|-------------|------------|---------|
| 101       | SQL basics  | IT         | 3       |
| 104       | Excel       | Finance    | 2       |
| 105       | Statistics  | HR         | 3       |

Question 8

```

SELECT Course_id, Course_name, Credits
FROM Course
ORDER BY Credits DESC
LIMIT 3;

```

| Course_id | Course_name  | department | Credits |
|-----------|--------------|------------|---------|
| 102       | Python       | IT         | 4       |
| 103       | Data Science | IT         | 4       |
| 101       | Equl basics  | IT         | 3       |



### Section 3

#### Question 9

```
SELECT min (grade) as min-grade,  
       Max (grade) as Max-grade,  
       Avg (grade) as Avg-grade,  
FROM   enrollments;
```

| Min-grade | Max-grade | Avg-grade |
|-----------|-----------|-----------|
| 78        | 80        | 84.6      |

#### Question 10

```
SELECT course_id, count(*) as enrollment-count  
FROM   enrollment  
GROUP BY course_id;
```

| Course_id | Enrollment-count |
|-----------|------------------|
| 101       | 1                |
| 102       | 1                |
| 103       | 1                |
| 104       | 1                |
| 105       | 1                |

### Section 4

#### Question 11

```
SELECT department, Sum (Salary) as total-salary,  
       Sum (bonus) as total-bonus  
FROM   Salaries  
GROUP BY department;
```



| Department | Total-Salary | total - bonus |
|------------|--------------|---------------|
| IT         | 68 000       | 10 500        |
| HR         | 109 000      | 7500          |
| Finance    | 70 000       | 6000          |

### Question 12

SELECT department, ~~avg~~ AVG (Salary) as Avg-Salary  
 FROM Salaries  
 WHERE Salary > 55 000  
 GROUP BY department;

| department | Avg-Salary |
|------------|------------|
| Finance    | 70 000     |
| IT         | 62 000     |

### Question 13

SELECT employee-id, Name, Salary, bonus,  
 (Salary + bonus) as total-compensation  
 FROM Salaries  
 WHERE (Salary + bonus) > 60 000;

| Employee-id | Name  | Salary | bonus | total-compensation |
|-------------|-------|--------|-------|--------------------|
| 3           | Epike | 70 000 | 6000  | 76 000             |
| 4           | Tyke  | 62 000 | 5500  | 67 500             |



## Section 05

### Question 14

```
SELECT department, Sum(budget) as total-budget,  
AVG(budget) as Avg-budget  
FROM Projects  
WHERE AVG(budget) > 70 000  
GROUP BY department;
```

| department | total-budget | Avg-budget |
|------------|--------------|------------|
| IT         | 270 000      | 135 000    |
| Finance    | 80 000       | 80 000     |

### Question 15

```
SELECT Project-id, Project-name, department,  
budget  
FROM Projects  
WHERE budget BETWEEN 50 000 and  
120 000 AND department <> 'marketing'
```

| Project-id | Project-name   | department | budget  |
|------------|----------------|------------|---------|
| 1          | AI App         | IT         | 120 000 |
| 2          | Payroll System | Finance    | 80 000  |
| 3          | Dashboard      | IT         | 150 000 |
| 5          | HR Payroll     | HR         | 50 000  |

